



A pragmatic approach to analyze the ability of ethylene-octene copolymer in the long chain branching of polypropylene in molten and solid state

Mohammadmahdi Negaresh¹ | Milad Karbalaee-Bagher¹ | Yousef Jahani² | Ali Forouzan²

¹Department of Polymer Engineering and Color Technology, Amirkabir University of Technology, Tehran, Iran

²Plastic Department, Faculty of Processing, Iran Polymer and Petrochemical Institute, Tehran, Iran

Correspondence

Mohammadmahdi Negaresh, Department of Polymer Engineering and Color Technology, Amirkabir University of Technology, Tehran, Iran.
Email: m_neg112@aut.ac.ir

Abstract

The investigation of chain branching in polypropylene (PP) holds significant importance for comprehending the impact of structural modifications on the properties of PP. The utilization of ethylene-octene copolymer (EOC) presents a valuable opportunity to examine the influence of branching density on the ability of electron beam irradiation or a chemical agent in a PP blend. The introducing long chain branches (LCB) into PP can enhance its mechanical properties, particularly impact strength. This characteristic is of particular significance in applications where the material is subjected to mechanical stress, such as automotive components or packaging materials. Blends of PP/EOC were produced using 20 and 30 wt% of EOC and 0.5 phr trimethylolpropane trimethacrylate (TMPTMA) monomer. To evaluate the efficiency of branching in both solid and molten states, irradiated samples and samples mixed with dicumyl peroxide (DCP) were selected. The rheological properties of the molten blends in shear and extensional modes were assessed, and their morphology was examined using scanning electron microscopy. The existence of LCB in all samples was confirmed through dynamic viscoelastic measurements. It was concluded that although the irradiation promoted chain scission within the backbone, which resulted in long chain branching, DCP created LCB on the backbone chains of PP through chemical reaction in the melt state. Additionally, the strain hardening constant (n) was calculated, and its value for the irradiated sample PP/EOC 80/20 was 0.19, whereas its value for this blend when employing DCP was 0.14. While the complex viscosity of irradiated blend (8764 Pa.s) was greater than that of melt state branched blend (8377 Pa.s) at 0.1 rad/s, in the higher frequencies it became smaller due to the more effective creation of long chain branches through peroxide grafting in the molten state. The results of the study verified the presence of LCB in the materials by observing longer relaxation time and strain hardening behavior.

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Highlights

- The chain branching of PP/EOC blends was investigated in both the melt state and solid state through the presence of TMPTMA monomer.
- The effectiveness of TMPTMA monomer in grafting was mostly notable in the molten state.
- The steady state viscosity of LCB-(PP/EOC) was higher in the presence of dicumyl peroxide compared to irradiated PP/EOC.
- The samples containing 80 wt% of PP exhibited longer side branches compared to the sample with 70 wt% of PP according to shear and extensional rheometry.
- The induced side chain branches resulting from irradiation and the use of dicumyl peroxide help increase the miscibility of the polymers to a similar extent.

KEYWORDS

ethylene-octene copolymer, irradiation, multifunctional monomer, peroxide, polypropylene

1 | INTRODUCTION

The polyolefins represent the foremost thermoplastics, commanding significant attention in various industrial sectors. Extensive endeavors are underway to expand their application across diverse branches of industry. The rapid advancement of polyolefins is attributed to their favorable physical and chemical attributes, economical pricing, and the ability to regulate molecular weight.¹ Notably, polypropylene (PP) stands out as a globally prominent and extensively employed polymer due to its exceptional physical properties, including rigidity, impact strength, and chemical resistance.^{2,3} As a commodity plastic characterized by a linear chain structure and a relatively narrow molecular weight distribution, polypropylene is primarily produced using Ziegler–Natta catalysts. It displays a lack of strain hardening behavior in its molten state, resulting in suboptimal processing characteristics under extensional flow. Consequently, its application in fields such as foaming, blow molding, extrusion coating, spinning, and thermoforming is limited.^{4–6} It is renowned as a key polymer for fiber formation, owing to its unique behavior during the melt spinning process.⁷

The processing procedures crucial for various applications are frequently characterized by elongational flow, significantly influencing operations such as fiber spinning, thermoforming, blow molding, film blowing, film casting, wire coating, and foaming when dealing with polymer melts.⁸ Hence, substantial endeavors have been dedicated to enhancing the melt strength and strain hardening characteristics of linear polypropylene through the grafting of long chain branches onto the polymer

backbone.^{9–11} The branching process primarily encompasses two key strategies: one involves the integration of units into linear PP via electron-beam treatment in the solid state,^{12–14} while the other is exclusively feasible in the molten state through chemical methods, such as the utilization of peroxides in conjunction with multifunctional monomers as coagents that possess relatively low decomposition temperatures.^{15–17}

Several methodologies leverage the direct synthesis of long-chain branched PP (LCB-PP): through the utilization of Metallocene or Ziegler–Natta catalysis (constrained geometry catalyst) directly^{18–20} or via the incorporation of pre-fabricated PP macromonomers²¹; by employing conjugated diene monomers^{22,23}; or by means of metallocene-mediated polymerization of PP in the presence of a T-reagent, specifically p-(3-butenyl) styrene.^{24,25} Diverging from alternative techniques, the two aforementioned strategies for generating LCB-PP in both the molten state and solid state have currently attained commercial viability.^{26–28} All of these methodologies yield LCB-PP with widened molecular weight distribution and intricate branched configurations. Notably, the length of lateral branches significantly influences the melt rheology and processing behavior of branched polymers, as well as the quantity, density, and microstructure of their branches.²⁹ Consequently, there has been a surge in the PP backbone as it is subjected to long chain branching, chain scission and crosslinking emerge prominently as concurrent side reactions.³⁰ Incorporating polyfunctional monomers enhances branching. Among these, trimethylolpropane trimethacrylate (TMPTMA) stands out as a highly effective multifunctional monomer that is

extensively employed to augment the efficiency of the branching process.³¹ In the study of Gotsis et al.¹³ various techniques were employed to indirectly assess the level of PP branching. The co-agents were ranked based on their capacity to promote long-chain branching (LCB) while minimizing cross-linking and degradation of iPP. Among them, furfuryl sulfide (FS) demonstrated the highest effectiveness in facilitating branching. On the other hand, divinylbenzene (DVB) was found to be unsuitable for promoting branching.

Tabatabaei et al.³² conducted research on the combination of linear and LCB-PP blends. They proposed that even a small incorporation of LCB promotes a higher concentration of nucleation sites, leading to an enhancement in crystallinity. Nevertheless, the introduction of additional branches restricts chain movement, consequently leading to a reduction in crystallinity (specifically observed when utilizing a 20% LCB-PP ratio within the blend). A successful branching reaction was achieved by utilizing peroxydicarbonates (PODIC), a unique class of peroxides, to generate PP with an LCB structure.³³ A later study extensively described that PODICs with lengthy aliphatic side chains (such as di-myristyl-peroxydicarbonate) resulted in a high branching efficiency. In this process, the PODIC acted as both an initiator and co-agent, leading to partial chain scission and radical stabilization, crucial for the successful branching reaction. Polypropylene branching is not always a desirable achievement. For instance, a research study aimed at enhancing the application of PP found that while only the LCB-PP exhibits strain hardening and possesses five times the melt strength of other grades, it does not offer the widest range for foaming or the highest foam quality in terms of density (140 g/L) and cellular morphology.³⁴

A remarkable technique for inducing long chain branching (LCB) in polypropylene (PP) involves the utilization of a copolymer such as ethylene-octene copolymer (EOC).³⁵ EOC is a terpolymer composed of ethylene, octene, and a comonomer, exhibiting excellent elastomeric properties and a significant degree of LCB.³⁶ Consequently, it effectively enhances the formation of LCB within the polymer structure. By incorporating EOC into PP through blending, a heterophasic structure is created, leading to notable modifications in the mechanical properties of PP, including increased stiffness and toughness.⁷ The introduction of LCB through EOC broadens the potential applications of PP, particularly in scenarios that necessitate enhanced impact resistance.³⁷ Moreover, irradiation has emerged as a popular and cost-effective method for inducing a more regular LCB in PP/EOC blends.³⁸ Through the irradiation process, free radicals are introduced, which then react with the EOC to facilitate a crosslinking reaction and subsequent formation of LCB.³⁹

Although irradiation can be conducted in various states (solid, molten, or solution), the solid state method is commonly preferred. Polypropylenes with LCB were synthesized through a melt reaction, combining a maleic anhydride functionalized polypropylene with various concentrations of glycerol.⁴⁰ The FTIR spectra of the resulting modified polymers exhibited a distinct band around 1735 cm^{-1} , suggesting the presence of ester groups. This confirmed the successful reaction between the hydroxyl (OH) groups of glycerol and the anhydrides of the maleic anhydride functionalized polypropylene (PPg).

A different synthesis method for producing long-chain branched polypropylene (LCB-PP) is reported.⁴¹ This method involves in situ ozonolysis during a reactive extrusion process with the presence of trimethylolpropane triacrylate (TMPTA) as a co-agent. Ozonation efficiently generates free radicals in molten PP, leading to chain scission reactions during extrusion. The addition of TMPTA allows for the formation of LCB structures in PP molecules. Higher concentrations of TMPTA result in an increased amount of LCB produced. The presence of long-chain branching significantly influences the viscoelastic properties of the melts. LCB-PP samples exhibit significantly higher zero shear viscosities compared to the original polypropylene. The onset of shear thinning occurs at lower frequencies, and the material gradually loses its Newtonian behavior with increasing TMPTA concentration.

The process of blending polypropylene with polymers having superior properties to reduce its weaknesses has been adopted in some cases. This approach aims to improve the overall performance of the resulting blend. For instance, the incorporation of PP with Poly(1-butene) (PB-1) has been revealed to yield enhancements in various properties, encompassing impact strength, flow characteristics, creep behavior, softness, ultimate elongation, tear strength, puncture resistance, and optical properties. Notably, PB-1 boasts a melt strength approximately twice that of PP, despite having a molecular weight that is largely similar.^{42,43} Historically, ethylene propylene diene (EPDM) and ethylene propylene rubber (EPR) have served as popular choices for fulfilling this requirement. However, metallocene-catalyzed ethylene/ α -olefin copolymers (ECs), notably octene-based ethylene/ α -olefin copolymers (EOC), have emerged as favored alternatives due to their distinct and advantageous properties when juxtaposed with other elastomeric polyolefins.⁴⁴ Also, the impact of EOC content on the crystallization behavior of the polypropylene (PP) phase was the subject of thorough investigation in PP/EOC blends.^{45,46} Through the application of dynamic mechanical thermal analysis (DMTA) and non-isothermal crystallization assessments, it was observed that the PP/EOC blend exhibits partial miscibility with minimal likelihood of co-crystallization,

primarily attributed to the substantial temperature disparity between PP and EOC. Furthermore, isothermal crystallization studies revealed that EOC exerts a decelerating effect on the crystallization rate of PP, owing to the formation of chain entanglements between PP and EOC. In another study, the inclusion of polypropylene-grafted maleic anhydride (PP-g-MA) as a compatibilizer, alongside nanoclay, was employed to modify PP/EOC blends featuring two distinct grades of EOC characterized by differing viscosities, utilizing a meticulous melt mixing process.⁴⁷ The primary objectives were to enhance the toughness of PP through the integration of elastomer and nanoclay. Notably, it was observed that the presence of the higher viscosity EOC grade within the PP/EOC blends engenders a droplet-matrix structure, characterized by larger droplets. However, this particular grade demonstrates superior dynamic mechanical properties, underscoring the dominant influence of viscosity. In the study of Entezam et al.,⁴⁸ the electron beam irradiation method at a relatively moderate irradiation dose of 20 kGy within an air atmosphere was applied to make blends of PP and EOC injectable. The determination of gel content and rheological characteristics brought to light the prevalence of long chain branching as the primary phenomenon engendered during the irradiation process of EOC.

The incorporation of the polypropylene/ethylene octene copolymer blend, irradiated in the presence of TMPTMA, holds great significance due to its immense potential in improving material properties. This study provides an in-depth exploration of the degree of branching in both the molten and solid states, offering a comprehensive analysis of the mechanism underlying the superior effectiveness of branching compared to previous research efforts. Additionally, the investigation into chain branching in polypropylene and ethylene-octene copolymers consistently demonstrates its impact on the regularity and mechanical characteristics of the resulting materials. Using a combination of rheology of solvent and anti-solvent methods with FT-IR measurements, this study carefully examines the influence of ethylene octene copolymer on long chain branching in the melt state in the presence of DCP. Furthermore, detailed analytical descriptions and comparisons of the grafting effect on rheological parameters contribute to a deeper understanding of this intricate subject matter.

2 | EXPERIMENTAL

2.1 | Materials

The polypropylene used in this study was the homopolymer Moplen HP500H from Arak Petrochemical Company, Iran.

The EOC (ENGAGE™ 8100 Polyolefin Elastomer) from Dow Company, with a density of 0.870 g/cm³, was utilized as pellets with a melt flow rate of 1 g/10 min (190°C; 2.16 kg). The Fourier transform infrared spectroscopy (FTIR) of EOC is depicted in Figure 1. The multifunctional coupling agent, trimethylolpropane trimethacrylate, with a density of about 1.06 g/cm³, was obtained from Sigma-Aldrich Company. Dicumyl peroxide (DCP) with a purity of 98% was purchased from Sigma-Aldrich Company and served as an initiator to generate radicals. Additionally, Irganox 1010 from BASF was used as an antioxidant.

2.2 | Sample preparation

2.2.1 | Electron-beam post-treatment

The PP and EOC were mixed with 0.75 phr of TMPTMA in a co-rotating twin-screw laboratory extruder from Brabender, Germany, with an L/D ratio of 540. The mixing was carried out at 60 rpm, and the temperature profile ranged from 175 to 195°C from the hopper to the die. The extrudates were cut into granules using a strand cutter. The mixed granules were then injection molded at a melt temperature of 185°C. The samples were irradiated using a high-power industrial electron beam accelerator of the Rhodotron type at the Yazd Radiation Processing Center in Iran. The energy of accelerated electrons was 10 MeV with a 4-mA beam current, which enhanced the penetration depth of electrons within the samples. The homogeneity of the absorbed dose in samples was measured by the dosimeter. The compounds were packed in two layers of polyethylene-coated aluminum bags with a thickness of 0.8 mm and vacuumed up to 99%. Finally, they were sealed after purging highly purified nitrogen gas (99.99%, HPLC grade) through a Henkeleman food-packaging device in Germany.

To investigate the impact of electron beam (EB) irradiation on the long-chain branching (LCB) process in ethylene-octene copolymers (EOC), it is crucial to minimize degradation reactions. To achieve this, the samples were exposed to a fixed dose of 12.5 kGy.^{49–51} After irradiation, the samples underwent the following annealing process:

1. Initial annealing under a nitrogen atmosphere for 30 min at 80°C to facilitate the mobility of free radicals.
2. Subsequent annealing for 90 min at 130°C to deactivate the remaining free radicals.

This annealing process helped to ensure that the EOC samples were properly prepared for further analysis and

FIGURE 1 FTIR Spectra of ethylene-octene copolymer.

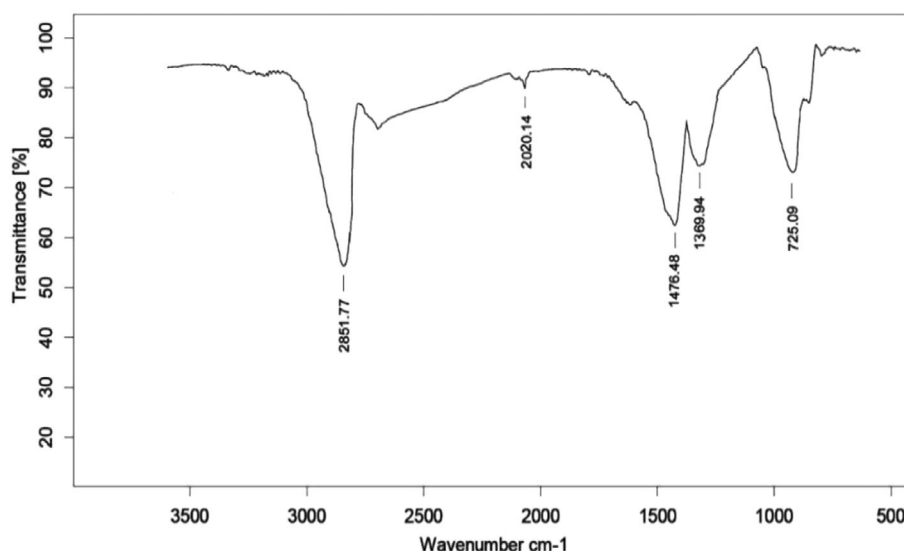


TABLE 1 Blend composition.

	PP (wt%)	EOC (wt%)	TMPTMA (phr)	DCP (phr)	Irad dose (kG)	IRGANOX 1010 Ppm
PP80	80	20	0.75	0	0	500
PP80-R	80	20	0.75	0	12.5	-
PP80-M	80	20	1.5	0.2	-	500
PP70	70	30	0.75	0	0	500
PP70-R	70	30	0.75	0	12.5	-
PP70-M	70	30	1.5	0.2	-	500

allowed for a more accurate assessment of the effects of EB irradiation on the LCB process.

2.2.2 | Melt reactive extrusion

The modified PP/EOC blends were prepared in two compositions (80/20 and 70/30) with a fixed amount of trifunctional monomer (1.5 phr) and dicumyl peroxide (0.2 phr) using a co-rotating twin-screw lab extruder with six heating zones (175–180–185–190–195–200°C). Following extrusion, the strands were immediately quenched in a water bath at room temperature and then chopped into granules. The detailed compositions of the blends are provided in Table 1.

3 | CHARACTERIZATION

3.1 | Morphological analysis

The morphology of all samples containing the irradiated and nonirradiated blends after processing was studied by a scanning electron microscope (SEM), (DSM 960 A,

Germany) operated at 10 kV accelerating voltage. The cryo-fractured samples in liquid nitrogen were kept immersed in cyclohexane to remove the EOC. After a specific time, samples were dried and gold-coated before the SEM test. The number and weight average diameter of 300 particles of dispersed phase were determined using the Image J analyzer.

The morphology of all samples containing the irradiated and non-irradiated blends after processing was analyzed using a scanning electron microscope (SEM) operated at 10 kV accelerating voltage (DSM 960 A, Germany). The cryo-fractured samples, immersed in liquid nitrogen, were treated with cyclohexane to remove the EOC. Subsequently, the samples were dried, gold-coated, and then subjected to the SEM test. The number and weight average diameter of 300 particles of the dispersed phase were determined using the Image J analyzer.

3.2 | Melting flow rate and gel fraction measurement

The flow properties of the modified blends were assessed using a melt flow apparatus from CEAST

(Modular line model, Italy) in accordance with the ASTM-D 1238 standard. The measurements were conducted at 230°C and with a 2.16 kg load. To determine the amount of the insoluble portion (macrogel) in the samples, a Soxhlet extraction was performed in refluxing xylene at 140°C for 12 h. Afterward, the residue was dried for 1 h at 150°C in a vacuum oven. Three different species were used to calculate the average gel content for each sample, following the ASTM D-2765 method.

3.3 | FTIR spectroscopy

The reacted samples were dissolved in boiling xylene at 130°C and then charged into methanol as an anti-solvent at room temperature. The process resulted in the precipitation of PP and PP-g-TMPTMA, while the unreacted TMPTMA monomer and homopolymerized TMPTMA remained soluble. The precipitated samples were filtered, dried at 60°C under vacuum for 12 h, and then mixed with Potassium Bromide (KBr) powder. The mixture was ground for 3–5 min and pressed at 8 MPa using a Qwik Handi-Press to create KBr pellets. The purpose of these pellets is to provide a thin, transparent sample for analysis, ensuring that IR radiation can pass through the sample for accurate assessment. Additionally, the use of KBr pellets helps achieve consistency in sample presentation, which is essential for obtaining reliable and reproducible results. The pelletized samples were analyzed using the Equinox 55 instrument (Bruker, Inc., USA) in the range of 4000 to 400 cm^{-1} with 64 scanning numbers. The resolution of the FTIR test was set at 8 cm^{-1} . In the results and discussion section, only the range 1000 to 1900 cm^{-1} was analyzed and reported due to its significance.

3.4 | Rheological measurement

Rheological measurements were conducted on various samples, including neat polymer and blends, using a Rheometrics Mechanical Spectrometer (RMS 800) with a 25 mm diameter disc obtained from 1.5 mm compression-molded sheets. The linear viscoelastic behavior of the samples was investigated in a frequency range of 0.01–600 rad/s with a strain amplitude of 3%. The linear viscoelastic range of deformation was characterized by a strain sweep test at a frequency of 10 rad/s at 180°C. Continuous nitrogen purging of the environment chamber was essential to minimize oxidative degradation of the blends. A time sweep test was carried out at a constant angular frequency of 1 rad/s, strain amplitude of 1%, and temperature of 180°C for most blends to ensure that degradation would not occur during the rheological test and the molecular structure

remained unchanged. The samples were also rheologically characterized in simple extension using a SER Universal Testing Platform from Xpansion instruments, equipped with a MCR301 rheometer and CTD450 convection oven (Anton Paar). The $15 \times 1.0 \times 0.5 \text{ mm}^3$ samples were prepared in a hot press at 180°C. Extensional viscosity was measured at constant Hencky strain rates of 0.03–1 s^{-1} at 180°C.

4 | RESULTS AND DISCUSSION

4.1 | Scanning electron microscopy

The SEM micrographs of all blends, before and after irradiation, are depicted in Figure 2A–F. It is evident that all blends exhibit a two-phase structure with a droplet-matrix morphology. The fine EOC droplets are homogeneously dispersed in the PP matrix, and the particle size diameter of the droplets and their structure remain unchanged after irradiation and the melt mixing process. The average radius of droplets (R_v) in all samples is approximately 30 nm. Similar findings were previously reported.⁵² The notable change is the impact of DCP on the miscibility of the polymers. The SEM micrographs reveal an increased mutual propensity of components, attributed to the higher induced side chain branches during melt state long chain branching (Figure 2C,D). Additionally, it is evident that the efficiency of irradiation is more pronounced in a blend containing 20 wt% of EOC. Similar results were obtained when DCP was used, although slight difference was observed in the case of irradiated samples regarding the lower miscibility of polymers.

The study revealed that the thermo-mechanical process of melt mixing induced the formation of PP macro-radicals. These reactive particles subsequently interacted with more stable EOC macro-radicals, resulting in the creation of EOC molecules with longer branches, particularly at the interface. These elongated EOC molecules played a crucial role in establishing a network of molecular interconnections, thereby facilitating the interconnectivity of EOC droplets across a broad range. Long side branches, which act as nucleating agents, promote the formation of more crystals at the interface of two polymer phases, leading to the entrapment of chains of two polymers in these areas.¹¹ Additionally, these long branches lead to more entanglement with the alongside main chains or chain branches, which can be another factor in enhancing the compatibility of polypropylene and ethylene-octene copolymer.

Microscope images at 100× magnification were used to precisely examine the impact of electron radiation on

FIGURE 2 SEM micrographs (30×) of: (A) PP80, (B) PP70, (C) PP80-M, (D) PP70-M, (E) PP80-R and (F) PP70-R.

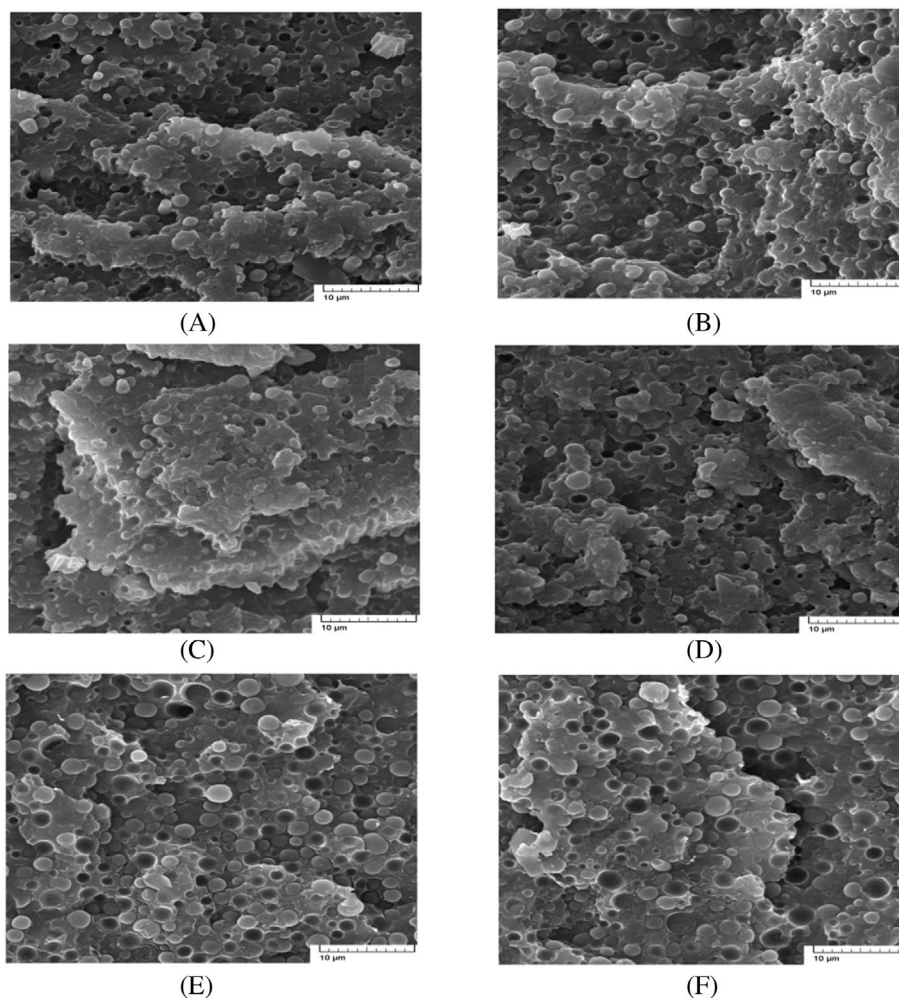
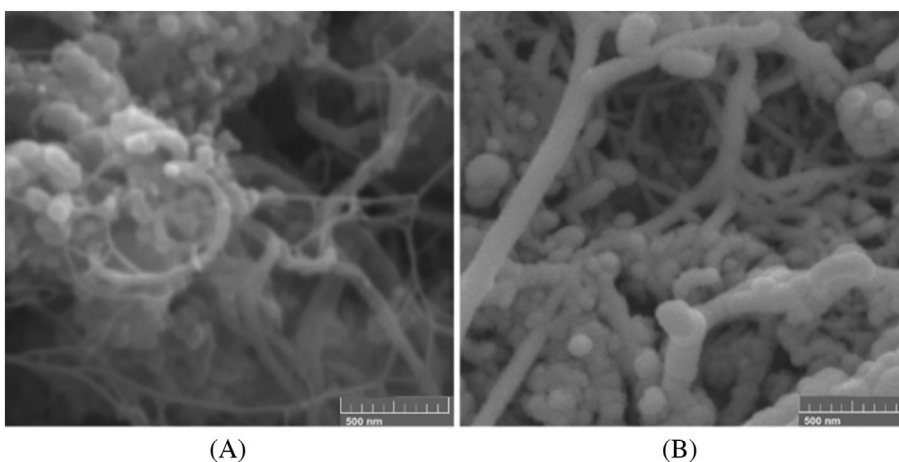


FIGURE 3 SEM micrographs (100×) of: (A) PP80-R, (B) PP80-M.



the miscibility of polymer phases. As shown in Figure 3B, sample PP80-M demonstrates an increased number of connections between polymer chains compared to PP80-R. This is attributed to the generation of a greater quantity of interpenetrating branches, consequently enhancing their compatibility.

4.2 | Fourier transform infrared spectroscopy

Figure 4 shows the FTIR spectra of purified samples. The spectra demonstrate that all of the TMPTMA-modified samples have bands at approximately 1730 cm^{-1} , which

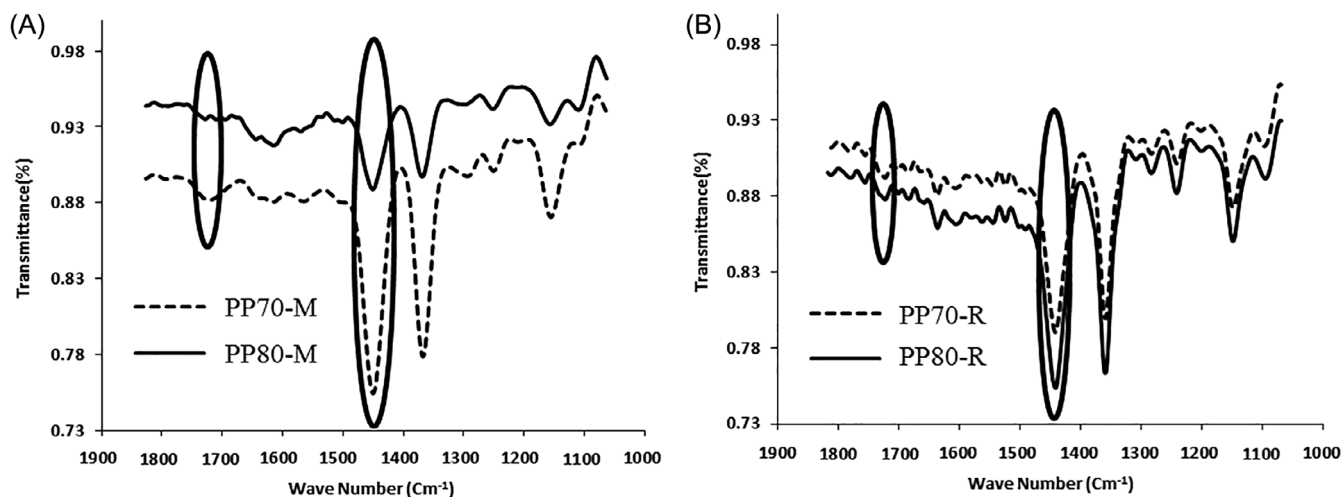


FIGURE 4 FTIR spectra (1000–1900 cm^{-1}) of purified samples: (A) for melt state branched, (B) for solid state branched.

are due to the carbonyl group stretching vibration of the ester in the TMPTMA structure. This indicates that the TMPTMA is grafted onto the PP or EOC backbone.

Various methods are available for analyzing the degree of branching in polymers, which can be used independently or in conjunction with other experiments to analyze the data. Some of methods of measuring grafting efficiency include analyzing the chemical composition distribution, crystallization kinetics study, molar mass characterization, and dynamic mechanical analysis.⁵³ In one of the recent studies, Hoffman–Lauritzen analysis of crystallization was performed for PP samples measured by optical microscopy.⁵⁴ Moreover, the Arrhenius equation was utilized, and the analysis involved conducting linear regression on its logarithmic graph to investigate the impact of electron radiation on the degree of crystallization of polypropylene and polyethylene samples.⁵⁵ In the context of exploring the impact of branching on molecular weight, Entezam et al.⁵⁶ elaborated on a comparison between the molar mass distributions (MMDs) for samples before and after irradiation indicates that changes in the size of HDPE chains due to the irradiation process mostly occur at the intermediate and high molar mass parts of the distribution. The carbonyl index (CI) is a suitable method for evaluating the degree of branching success in polymer blends, and can be easily coupled with FTIR results.⁵⁷ Overall, molar mass characterization can reveal the overall molecular weight distribution, while CI specifically targets the presence of carbonyl groups as a measure of grafting efficiency.⁵⁸

The Carbonyl Index is a method used to measure how well side chains are attached to a polymer through reactive processing. The higher the CI value, the higher the degree of attachment, which is also known as grafting. This method is useful for evaluating the effectiveness of

purification methods and verifying the success of the grafting process.⁵⁹ The CI values of grafted samples can provide insights into the extent of functionalization and the resulting properties of the modified polypropylene. It is important to note that CI specifically focuses on the presence of carbonyl groups, which are indicative of the grafted maleic anhydride.

The CI is a relative value that measures the degree of attachment in a compound. It is calculated by dividing the under-curved area of the carbonyl peak ($\text{C}=\text{O}$) (A_{1730}) by the area of the methylene ($-\text{CH}_2$) scissoring peak from the absorbance at 1460 cm^{-1} (A_{1460}) (Equation (1)):

$$\text{CI} = \frac{A_{1730}}{A_{1460}} \quad (1)$$

A_{1460} is an internal reference band that considers the effect of grafting onto the EOC backbone, which contributes to the CH_2 groups in both PP and EOC backbone. A higher CI value indicates a higher degree of attachment. To create a calibration curve for measuring the grafting degree of samples, blends with specified amounts of TMPTMA and PP/EOC ratio are prepared using an internal mixer. This calibration curve allows the research to examine the grafting degree of samples more accurately (Figure 5). Tian et al.⁶⁰ employed a similar method to calculate the degree of branching for PETA-g-PP.

During the branching process, side reactions such as β -scission and homopolymerization of unsaturated monomers can occur randomly. β -scission leads to the degradation of PP and EOC chains, while homopolymerization can result in the formation of short chain structures and even large gels.⁶¹ According to the gel fraction

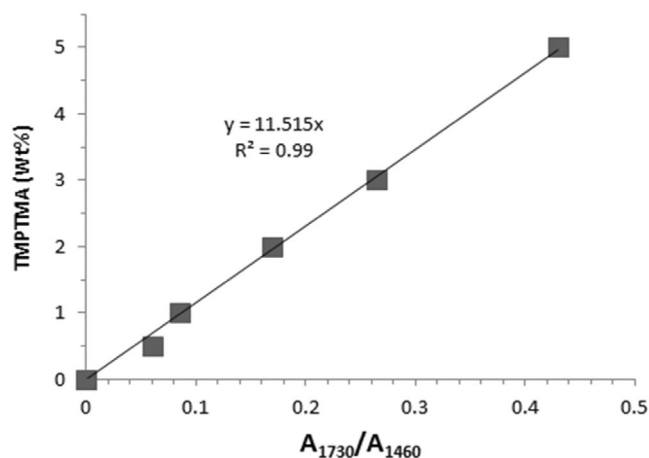


FIGURE 5 Calibration curve for the investigation of the percentage of reacted TMPTMA in PP-g-TMPTMA.

experiment, no gel in the modified samples was observed, which revealed no crosslinking entity throughout the branching process. The ability of EOC to form a network was influenced by its blending with polypropylene and the presence of the branching agent.⁶² As a result, the branching outcompeted crosslinking formation, leading to a minimal amount of gel in the sample representing networking.

The electron-beam treatment occurred exclusively within the amorphous region of the blends, which is why all parts of the modified samples underwent melt branching in an extruder. Consequently, the following observations can be made:

1. The degree of grafting in solid-state branching was lower than in melt-state branching.
2. As the amount of EOC increased in the solid-state, the grafting degree decreased.
3. Adding 30 wt% of EOC instead of 20 wt% in the melt state led to a significant enhancement in the grafting degree.

The efficiency of the polyfunctional monomer is higher in the melt state compared to the solid state due to factors such as freedom of movement, reduced chain entanglement, improved solvent action, and reduced degradation. In the melt state, polymer chains have more freedom to move and rearrange themselves, which can facilitate the grafting process. In the solid state, the polymer chains are more restricted in their movement, which may hinder the grafting process. In the melt state, polymer chains are less likely to be entangled with each other, making it easier for the grafting process to occur. In the solid state, polymer chains are more likely to be entangled, which can

impede the grafting process. In grafting, the use of a solvent can help to swell the polymer and make it more receptive to the grafting process. In the melt state, the solvent can be more effectively distributed throughout the polymer, enhancing the grafting efficiency. Grafting in the melt state can result in less degradation of the polymer compared to the solid state. This is because the melt state allows for more efficient grafting, which can lead to better bonding and fewer weak spots in the final product. The melt state can be more compatible with certain solvents and grafting techniques, which can lead to better results compared to the solid state. For example, reactive extrusion grafting can be conducted in the melt state, which can result in high grafting efficiencies.

4.3 | Shear rheology

Long chain branching is an important feature in polymer science that influences the rheological properties of polymers. LCB enhances polymer chain entanglements, which reduces polymer hydrodynamic volume and increases the viscosity of the polymer at low shear rates.⁶³ However, LCB polymers are more prone to be disentangled than linear ones due to their small hydrodynamic volume. It has been shown that LCB causes strain-hardening behavior in the extensional flow of polymers, which is not seen in linear species. The rheological study of LCB polymers has shown an increase in melt shear and elongational strengths, as well as an increase in shear thinning and strain hardening behavior.⁴⁴ The long-chain branching structure promotes the formation of crystals and reduces the final size of crystals. The chain length and branching degree strongly affect the chain motion in the polymer melt and determine the rheological behavior of the polymer. Therefore, LCB has a significant impact on the rheological behavior of polymers, affecting their viscosity and flow curves.⁶³

The linear viscoelastic behavior of molten polymers is highly dependent on molecular structure, and established experiments in the linear regimes can provide valuable information about long-chain branching. However, the more elaborate techniques, such as GPC and NMR, can produce challenging results that require clarification through nonlinear experiments. Shear rheological measurements alone can be ambiguous, so they should be accompanied by nonlinear experiments. In the Newtonian region at low shear rates, the zero-shear viscosity is known to depend on molecular weight, and long-chain branching is described by Equation (2):

$$\eta_0(M, \text{LCB}) = [k(gM)^a] \Gamma(\text{LCB}) \quad (2)$$

where the Mark-Houwink equation constants k and a , the viscosity enhancement factor Γ due to long-chain branching, and the variable g defined according to an equation are related to polymer solutions (3):

$$g = \frac{\langle S^2 \rangle_B}{\langle S^2 \rangle_L} \quad (3)$$

where S is the radius of gyration, the subscripts B and L designate “branched” and “linear” respectively.^{64,65} Long-chain branching (LCB) on the PP backbone has two well-established effects. First, it leads to a decrease in the radius of gyration, resulting in fewer entanglements and lower viscosity. Second, when the branch length is longer than twice the entanglement length in the melt, it leads to viscosity enhancement.^{66,67} Figure 6 presents the flow curves reduced by $a_M = [\eta_0(M)/\eta_0(M_{\text{ref}})]$ versus shifted frequency for each sample. To study the contribution of LCB to complex viscosity, viscosity flow curves were normalized to M_w and shifted in accordance with a reference molecular weight. The most logical reference materials, unmodified resins such as PP70 and PP80, were selected because both of them are linear.⁶⁸ The resulting flow curves were reduced by $a_M = [\eta_0(M)/\eta_0(M_{\text{ref}})]$ versus shifted frequency for each sample, providing a clear picture of the contribution of LCB to complex viscosity.

The modified samples in two blend ratios exhibit an increase in zero-shear viscosity compared to linear ones, indicating the formation of long-chain branched polypropylenes (LCB-PPs) during the process. This results in a reduction of the Newtonian plateau and less intensive pseudoplastic behavior, along with a broadened Power-law region.^{69–71} The irradiated samples also demonstrate higher viscosity than melt-branched ones, possibly due to increased LCB formation. However, definitive

conclusions cannot be drawn without considering nonlinear deformation.

Moreover, the sensitivity of dynamic modulus to sparse long-chain branching is evident in the behavior of the modified PP samples. While neat PP, PP80, and PP70 display typical terminal flow behavior, the modified PP samples deviate from this behavior. The branched polymers exhibit enhanced values of G' at low frequencies and a broader transition between the terminal zone (low frequencies) and the plateau zone (high frequencies), indicating a new, well-defined relaxation regime (Figure 7). This behavior suggests the presence of a distinct relaxation mechanism.^{72–74}

The interfacial tension of PP/EOC blend was found to be significantly higher (five times higher than for the LCB-(PP/EOC) resulting in extremely lower adhesion between phases. It indicates lower bonding and a negligible role of the interphase in the properties of this material.⁷⁵

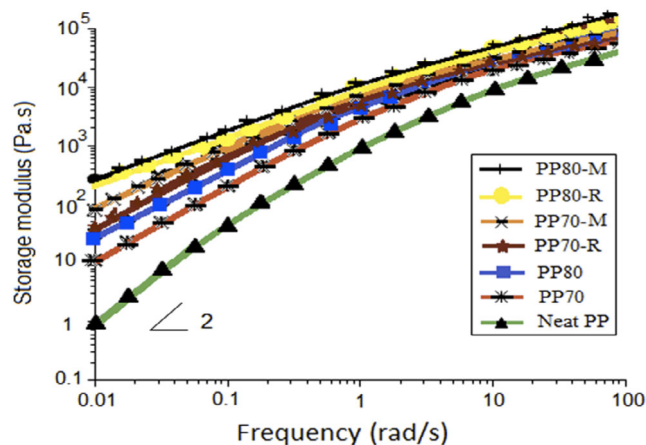


FIGURE 7 Storage modulus of the branched and unmodified samples against frequency at 180°C.

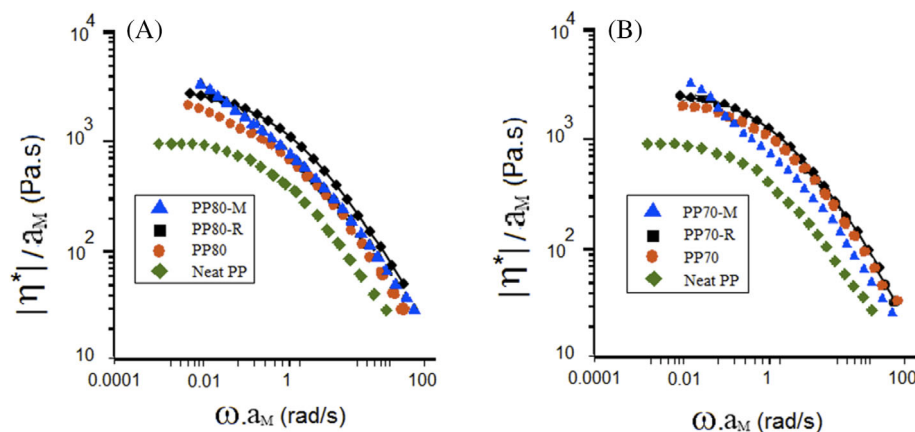


FIGURE 6 Plots of shifted complex viscosity versus frequency for: (A) PP80, and (B) PP70 sample series at 180°C.

Long-chain branching generally increases the viscosity of polymers. The presence of side chains introduces additional entanglements between polymer chains, leading to increased resistance to flow. This increase in viscosity is more prominent at lower shear rates. Polymers with long-chain branching often exhibit shear thinning behavior, which means that their viscosity decreases with increasing shear rate. This is due to the stretching and alignment of polymer chains along the flow direction, reducing entanglements and facilitating flow. Long-chain branching can also enhance the elastic properties of polymers. The presence of branches restricts the chain mobility, leading to higher elasticity and the ability to recover from deformation. This elastic behavior is evident in the flow curves as a plateau region at low shear rates, indicating the dominance of elastic response over viscous flow. According to Figure 8, the PP70-R sample exhibits additional elastic response due to the interconnectivity of EOC droplets caused by the bridging of grafted long chain-branched molecules and the intermolecular entanglements formed between long chain-branched and PP chains. While the complex viscosity of PP80-M (8377 Pa.s) is lower than that of PP80-R (8764 Pa.s) at 0.1 rad/s, it becomes greater in the higher frequencies become due to the more effective creation of long-chain branches through peroxide grafting in the molten state. The viscosity drop rate (n) is more gentle for PP70-M than for PP70-R, indicating a higher relaxation time for PP70-M, which is reflected in the viscosity changes in the frequency range of 10 to almost 100 (Figure 8). In another study, the use of LCB PPs in a blend resulted in higher viscosity at lower frequencies, while the incorporation of linear homo PP into the same blend material showed a decrease.⁷⁶

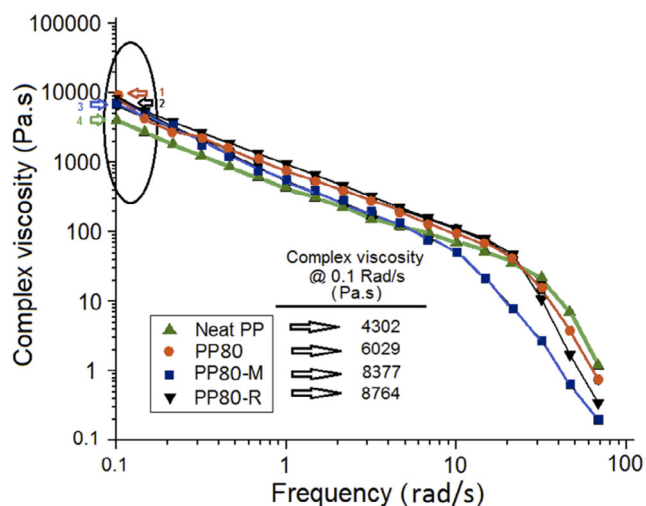


FIGURE 8 Complex viscosity of the branched and unmodified samples against frequency at 180°C.

4.4 | Extensional rheometry

The presence of side branches in polymers significantly affects their strain hardening behavior. Research indicates that branching enhances strain hardening in polymeric materials. Branches hinder the elongation and relaxation of polymer chains, resulting in an increased resistance to extension. For example, a study on polyolefin melts in extensional flows found that the stress resistance to uniaxial extension is closely related to chain length and the degree of branching.⁷⁷ Long chain branching has been shown to cause strain hardening behavior in the extensional flow of polymers, a feature not seen in linear species. Moreover, the specific number of branches per molecule has been found to affect strain hardening, with a higher number of branches leading to larger strain hardening and a more pronounced dependence on elongational rate.⁷⁸

The nonlinear behavior in extensional deformations provides valuable insights into the structural features of molecules that are not revealed by shear data, unlike linear viscoelastic data. Specifically, long-chain branching (LCB) significantly affects the melt response to a stretching flow, leading to the strain-hardening phenomenon.⁷⁹ When the nonlinear data exceeds the linear curve at certain points, the melt is termed strain hardening, caused by the restricted stretching of the backbone between the branch points connecting the branches.^{52,53} This behavior is observed in all modified samples at all Hencky strain rates, as depicted in Figure 9. A similar study by Ahul et al.^{10,54} also refers to the slightly branched PP.

As can be seen in Figure 9, all samples have the strain hardening behavior. This is because of the presence of side branches in polymers, which can restrict the movement of polymer chains when the material is subjected to strain or deformation. These branches act as physical obstacles that hinder the movement of chains, making it more difficult for the polymer to deform or flow. As a result, PP70-M with more side branches tends to exhibit higher degrees of strain hardening. The presence of these branches increases the resistance to deformation, requiring a higher level of stress or strain to continue the deformation process. Additionally, the side branches can also contribute to an increased elastic recovery in the polymer. Elastic recovery refers to the ability of a material to return to its original shape after deformation. The presence of side branches can enhance the elasticity of the polymer, allowing it to recover its shape more efficiently. The relationship between strain hardening and the presence of side branches in polymers is thus that the branches restrict the movement of polymer chains and increase the resistance to deformation, resulting in a higher degree of strain hardening.

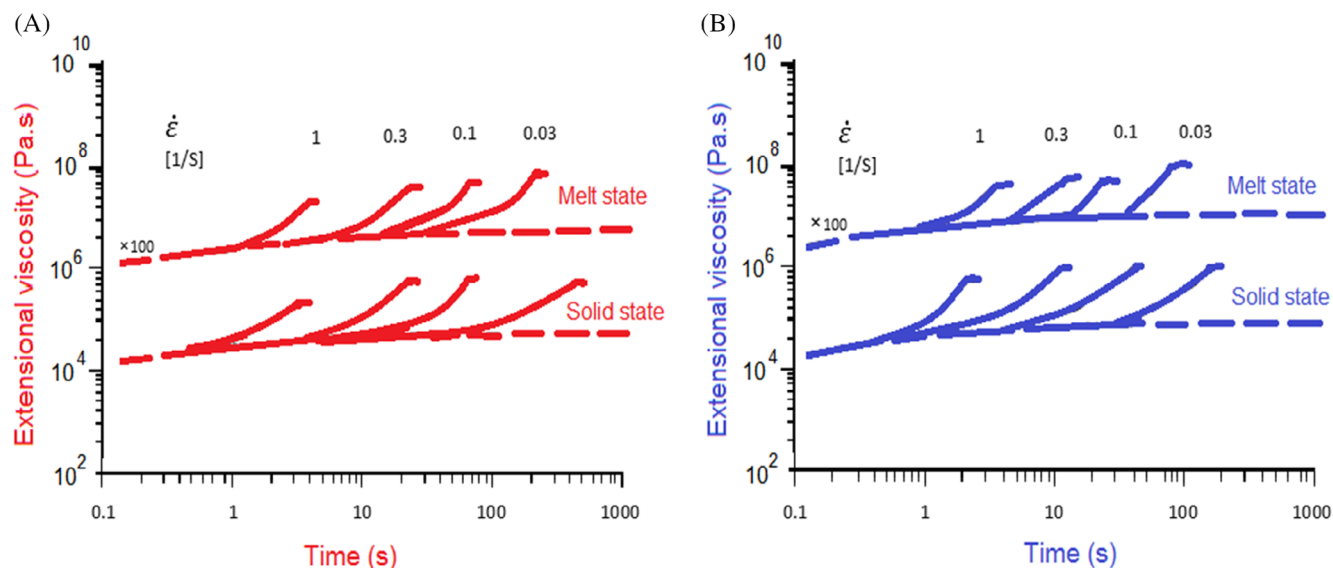


FIGURE 9 Time-dependent extensional viscosity at different Hencky strain rates for modified samples: (A) 80/20 and (B) 70/30 blend composition at 180°C.

The molecular stress function (MSF) theory is a model used in rheology to predict the viscoelastic stresses of polymeric fluids under given deformation histories. It is an integral constitutive equation that has been in use for more than two decades, which is based on an energy balance approach. This allows MSF to capture the complex structure of polydisperse, and randomly branched polymer systems. However, the quadratic version of the molecular stress function (QMSF) theory is more useful in this study because it is a constitutive model that describes the relationship between the stress and deformation of a material. The QMSF theory is used to predict the behavior of complex fluids and polymers under different flow conditions. In the QMSF theory, first proposed by Wagner et al.,⁸⁰ the added parameter, β , to the equation was used to separate the effect of elongation of the backbone and the compression of the side chains onto it. This parameter also determines the slope of strain hardening, and its value shows the number of side branches per molecule. Accordingly, in this work, the onset of strain hardening was utilized as a representative of the number of side branches to compare irradiated and melts branched PP. The slope of uprising extensional viscosities is plotted versus various Hencky strain rates in Figure 10. In general, as the extension rate increases, an ascending trend for the slope is observed. A careful look at the curves shows that the PP80-M has the highest slope, whereas PP70-R is placed below all other samples with the least number of branches. The results are approved by measured grafting degree parameters in the previous section. Even though grafting parameters for PP80-R and PP70-R were lower than PP80-M therein, they exhibit a

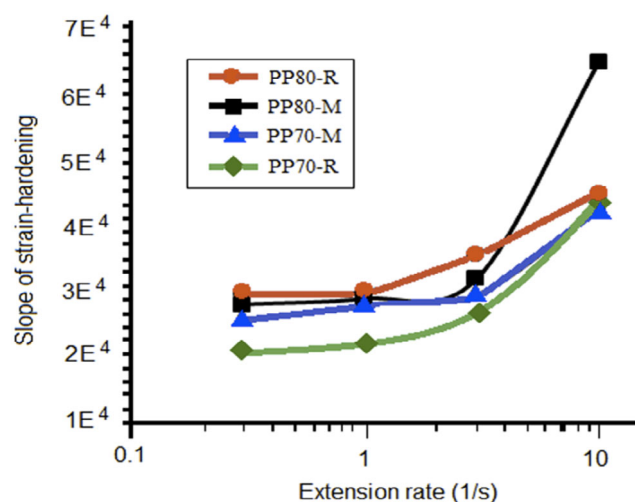


FIGURE 10 The slope of strain hardening versus various deformation rates for branched samples.

steeper slope. It is borne in mind that solvent, anti-solvent methodologies take advantage of how much trifunctional monomers were grafted onto PP backbones with the probability of not creating side branches or only one or two functional groups get involved in the reaction.

Commonly, in the process of determining material strength coefficient and strain hardening constant, material testing is initially conducted to obtain engineering stress and engineering strain, which are then converted to true stress (σ) and true strain (ϵ).⁸¹ Subsequently, these values are used in the flow stress formula, which can be expressed as $\sigma = K\epsilon^n$, where K represents the material strength coefficient and n represents the strain hardening

constant.⁸² However, obtaining these parameters involves complex and lengthy calculations.

The strain hardening constant (n) in the axial direction was calculated to evaluate the branching in different samples, and the results were presented in Figure 11. In the melt-state branched sample (PP80-M), the value of n is the lowest, while the elongation at break is higher. These findings are consistent with data reported in other articles.⁸³ Notably, pure polypropylene exhibits a higher true tensile strain at break but a lower n constant compared to all samples. Furthermore, as a result of electron beam radiation, the value of n decreases to 0.19, which is higher than that of PP80-M, equal to 0.14. This decrease in the value of n observed in branched samples indicates a significant impact of long branches under these conditions.

The steady-state extensional viscosity, a nonlinear parameter in the molecular stress function (MSF) theory, is denoted by $f_{\max}^2$ and defines the length of side branches in extensional flows. This parameter governs the steady-state value of viscosity in extensional flows and depends on the molecular topology.⁸⁴ A brief physical concept of these two parameters was schematically illustrated in Figure 12. The nonlinear parameter is associated with the transient and steady state uniaxial viscosities, and it is essential for predicting the rheological

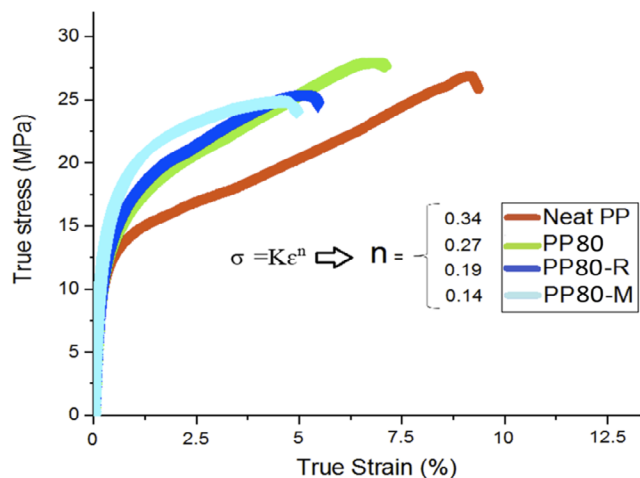
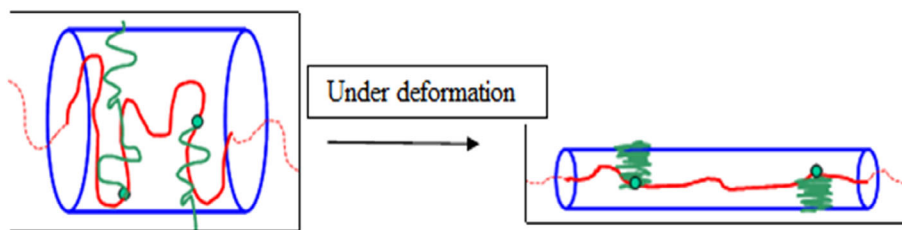


FIGURE 11 True stress versus true strain plots of PP, PP80, PP80-M, and PP80-R (in axial direction).

FIGURE 12 Tube segment of a long-chain branched polymer molecule before and after deformation. Side chains are compressed perpendicular to the backbone, while the backbone is stretched along the tube.



properties of various materials, such as polymers and polymer nanocomposites. The nonlinear parameter is influenced by factors like molecular topology, presence of particle networks, and the contribution of polymeric chains.

Given the aforementioned statement, we have employed the absolute values of maximum extensional viscosities pertaining to varying extension rates in Figure 13. These values are delineated in Table 2.

The higher rates of extension result in reduced values for the steady-state extensional viscosity due to the simultaneous unentanglement of chains. Conversely, at lower rates, all chains undergo uncoiling before experiencing pressure to free themselves from lateral entanglements, leading to a much greater stretching effect.

As anticipated, the PP80-M sample exhibits the longest side branches compared to the irradiated polymers. An EOC content exceeding 20 wt% causes shorter chains as C_2H_5 is more susceptible to degradation than PP80.

A comprehensive examination and analysis of CI, steady shear viscosity, melt flow index (MFI), and the terminal slope of storage modulus versus frequency can enhance our understanding of the impacts of chain branching and grafting efficiency on the rheological and flow properties of polymers. For instance, changes in CI

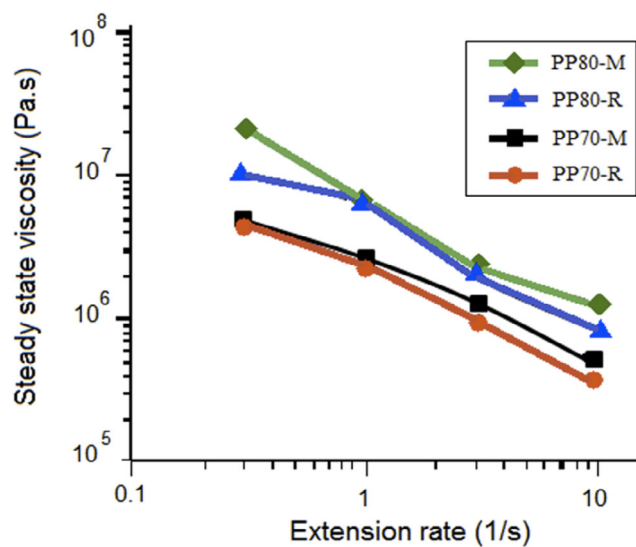


FIGURE 13 Steady state extensional viscosity versus different Hencky strain rates for branched samples.

	CI	S.S Viscosity (Pa.s)	MFI (g/10 min)	Terminal slope
PP80	-	-	1.9	1.15
PP80-M	28	9.81E+06	1.1	0.6
PP80-R	18	4.59E+06	6.4	0.86
PP70	-	-	1.9	1.3
PP70-M	21	4.64E+06	1.3	0.65
PP70-R	14	1.98E+06	8.7	0.95

TABLE 2 Characteristic features of materials.

may correlate with alterations in steady shear viscosity and MFI, indicating the influence of grafting on the flow behavior and viscoelastic properties of polymers.⁴⁷ Furthermore, the terminal slope can provide further insights into the viscoelastic response of the grafted polymer under flow conditions, contributing to a better comprehension of its behavior. All results are outlined in Table 2.

The analysis demonstrated that the melt flow index (MFI) of PP80-M was lower than that of PP, with its terminal slope reduced by half. In contrast, the MFI of PP80-R, which underwent electron beam irradiation, increased significantly, nearly six times that of PP80-M, and exhibited an approximately 43% increase in the terminal slope. This difference in viscosity and MFI is attributed to the formation of stronger and more numerous branches, as well as extensive inter-chain connections following melt-state branching. This relationship is directly connected to the degree of branching and the carbonyl parameter, rendering the flowability of samples more challenging to shear and extension stress. Furthermore, increasing the ethylene-octene copolymer content to 30 wt% resulted in an increase in MFI for all branched or unbranched samples compared to the blend containing 20% of this component. The decrease in viscosity and carbonyl index, almost by half, can be attributed to the negative effect of the ethylene-octene copolymer on the efficiency of the branching agent, as well as the reduction in the degree of electron penetration to polypropylene as the main branching part. Stanic et al.⁷⁶ utilized the incorporation of dilauroyl peroxide-based blends of LCB-PP with two peroxide modifications to determine whether it could enhance the aspect of melt strength behavior. It was shown that, with the introduction of LCB on PP chains, as the MFI values increased, the curves and the intersection points of G' and G'' of the samples were shifted toward higher frequencies.

Increased long chain branching results in a higher MFR, meaning the polymer will have more fluidity during processing. This allows for easier flow through machines, like extruders or injection molding equipment,

facilitating faster production rates. On the other hand, excessive long chain branching can decrease the melt strength of the polymer, resulting in poor processability. This can lead to challenges in various processing techniques, such as blow molding or film extrusion, where sufficient melt strength is required for optimum production efficiency.

The observation that grafting mostly occurred in the melt state can be attributed to a few factors. In the melt state, the polymer chains have higher molecular mobility compared to the solid state. This increased mobility allows the polyfunctional monomers to diffuse and react with the polymer chains more effectively, leading to a higher likelihood of grafting. In addition, the polyfunctional monomers are more likely to diffuse into the polymer matrix in the melt state due to the absence of a rigid structure. This promotes intermolecular mixing and increases the chance of encountering reactive polymer chains for grafting. In the solid state, intermolecular forces or crystalline structures, making it more challenging for the polyfunctional monomers to access and react with these sites, may hinder the reactive sites on the polymer chains. In the melt state, however, these hindrances are minimized, allowing for easier access to the reactive sites. On the other hand, electron irradiation provides energy that can break chemical bonds and generate reactive species. In the melt state, the polyfunctional monomers can more easily react with the generated reactive species due to their increased mobility. This leads to a higher grafting efficiency compared to the solid state. The melt state allows for a more uniform distribution of the polyfunctional monomers within the polymer matrix. This homogeneity ensures a higher probability of encountering reactive sites on the polymer chains during electron irradiation, further improving grafting efficiency. The melt state facilitates mass transfer since the polymer chains are in a molten state. This enhances the diffusion of free radicals or excited species generated by electron irradiation, enabling them to interact more readily with the polyfunctional monomers and react with the polymer chains. In the context of the importance of creating side branches in linear polymers, it should be noted that

polymer blends with appropriate long chain branching can be used for film extrusion applications. The presence of long chain branches can improve the optical properties of films, like transparency and gloss, while maintaining good mechanical strength and flexibility. Also, copolymer blends with well-controlled long chain branching can be used in injection molding applications. The presence of long chain branches can facilitate better melt flow, enabling the molds to be filled more easily and ensuring consistent part quality. Moreover, the impact of long chain branching on blow molding applications is highly dependent on the specific requirements of the end product. Controlled long chain branching can provide improved melt strength, leading to better parison control and blow molding performance for applications such as bottles or containers.

5 | CONCLUSION

Different blends of polypropylene/ethylene-octene copolymer (PP/EOC) and their modified counterparts were prepared using both melt state and solid state branching techniques. Rheological investigations were conducted on all samples to compare the number and length of side chain branches. The effectiveness of a polyfunctional monomer in the branched samples was measured using solvent and anti-solvent methodology, revealing that grafting mostly occurred in the melt state. This can be attributed to increased molecular mobility, better diffusion, enhanced access to reactive sites, and improved reaction kinetics in the melt state, which contribute to higher grafting efficiency compared to the solid state with electron irradiation. Furthermore, it was observed that an EOC content above 20 wt% had a negative impact on grafting efficiency during the two branching processes. However, our findings from extensional rheometry indicated that the PP/EOC (70/30) blend containing dicumyl peroxide had a higher number of formed side branches compared to the electron irradiated PP/EOC (70/30) blend, as evidenced by the terminal slope of G' versus frequency being 0.6 and 0.86, and the MFI being 1.1 and 6.4 (g/10 min), respectively. Increased long chain branching results in a higher MFR, meaning the polymer will have more fluidity during processing. Moreover, to eliminate the effect of chain entanglement that could influence the sheer effect of long chain branching, the shear viscosity was normalized to the molecular weight of the samples. Among them, the irradiated PP/EOC (80/20) and PP/EOC (70/30) blends demonstrated lower steady state viscosities ($4.59\text{E}+06$ and $1.98\text{E}+06$ Pa.s, respectively). This is why the lack of information in the shear data was revealed by the

extensional measurements, which showed that both the PP/EOC (80/20) and PP/EOC (70/30) blends containing dicumyl peroxide exhibited greater side branch length and number.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in [repository name e.g., “figshare”] at <http://doi.org/10.1002/pls2.10115>, reference number [81].

ORCID

Mohammadmahdi Negaresh  <https://orcid.org/0009-0003-6239-9571>

Yousef Jahani  <https://orcid.org/0000-0001-5553-1915>

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